Agentic Search Efficiency for Scientific Data Discovery

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Abstract

Scientific data search demands reasoning that does not scale: inspecting what each dataset contains, converting between unit conventions, refining queries when initial results fall short. AI agents can automate this, but the cost of running them is rarely examined. A single query can consume nearly a million tokens before producing an answer.

We present CLIO Search, which implements an observe–decide–act–evaluate cycle for scientific data. Science-aware operators execute deterministic SI conversion and formula normalization alongside BM25 and dense search. CLIO inspects what metadata each corpus provides and adapts accordingly. Federated search spans heterogeneous HPC backends with agentic refinement.

We compare three agentic paths over 10 discovery queries against the National Data Platform (NDP, 341 datasets from NOAA, NASA, ECMWF, and others) using `claude-sonnet-4-20250514`. An agent backed by CLIO reduces tokens by **52%** and wall time by **91%** relative to a native Claude agent, while answering all 10 queries correctly.

CLIO Search

CLIO Search implements an **observe–decide–act–evaluate** cycle for scientific data discovery:

1. **Discover (Observe):** Profile each corpus—document count, measurement count, metadata density, unit domains—to understand what search strategies are viable.
2. **Reason (Decide):** Select retrieval branches based on what the corpus provides. Skip the scientific branch if no measurements exist; skip vectors if no embeddings are indexed.
3. **Transform & Search (Act):** Execute four parallel branches—**BM25**, **dense vector** (MiniLM-L6-v2), **graph traversal**, and **scientific operators** (SI conversion across 58 units in 12 domains + formula normalization)—then merge with a scientific hard filter.
4. **Execute (Evaluate):** Assess results and decide to **expand**, **narrow**, **pivot**, or **stop**. Rewritten queries re-enter the pipeline.

The agent receives compact ranked citations—not raw catalog JSON. CLIO handles unit conversion, profiling, and ranking *before* the agent sees the results.

System Architecture

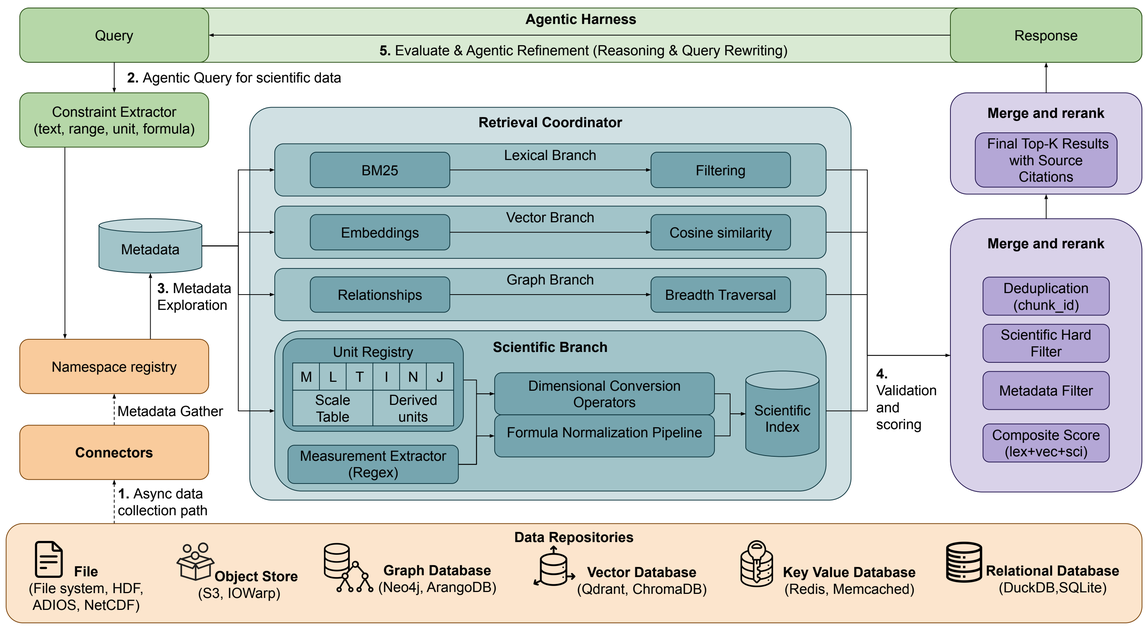

Figure 1: CLIO Search architecture. Connectors index data from heterogeneous repositories (filesystems, HDF5, S3, IOWarp, NDP). At query time, the Namespace Registry routes queries to capable branches; the Retrieval Coordinator executes lexical, vector, graph, and scientific branches in parallel; Merge & Rerank enforces dimensional correctness and returns top-K results. The agentic loop refines queries iteratively.

Acknowledgments

This work is part of the CLIO Kit project within the IOWarp platform. CLIO Search is open source at `github.com/SIslamMun/clio-search`.

Three-Path Experiment

We compare three ways of answering the same **10 scientific discovery queries** (temperature, pressure, wind, radiation, soil moisture) against the **National Data Platform** (NDP, 341 datasets from NOAA, NASA, ECMWF, and others), using `claude-sonnet-4-20250514` as the LLM:

Path 1 — Claude Native Agent: Web search, file tools, and sub-agent spawning. No domain-specific search. Must find NDP datasets by searching the open web.

Path 2 — NDP-MCP: Catalog search via Model Context Protocol. Returns *raw dataset JSON*—the agent must parse and re-rank in context.

Path 3 — CLIO: Agent queries CLIO Search, which has indexed and profiled all 341 NDP datasets. Returns compact *ranked citations* (∼2K tokens per query).

Results: Same Accuracy, Dramatically Different Cost

All three paths answer **10/10 queries correctly**:

Method	Tokens	Token △	Time △	Correct
Claude Native	989,417	—	—	10/10
NDP-MCP	988,954	−0.05%	−65.7%	10/10
CLIO	**474,101**	**−52.1%**	**−91.3%**	**10/10**

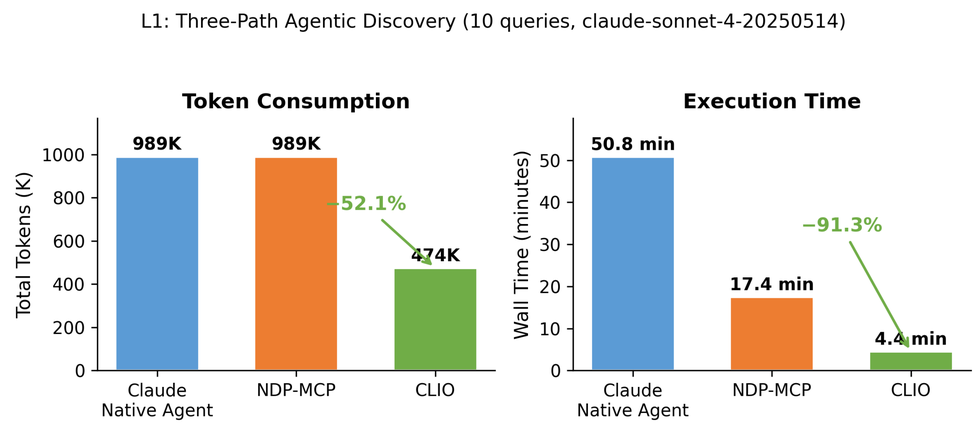

Figure 2: Token consumption (left) and wall time (right) across 10 queries. CLIO achieves 52% token and 91% time reduction.

Why CLIO Is Faster

1. **Smaller context:** CLIO returns ∼2K tokens of ranked citations per query versus 989K tokens of raw JSON. The native agent spawns 9 sub-agents (563K + 426K tokens); NDP-MCP floods the context with raw catalog JSON.
2. **Fewer reasoning steps:** 9/10 queries resolve in a *single invocation*. Corpus profiling, unit conversion, and ranking execute as deterministic local operations—the agent interprets results rather than navigating raw data.

Key Insight

The bottleneck in agentic search cost is not the number of tool calls but the **shape of the data** the agent sees. Compact, pre-ranked citations scale better than raw catalog JSON. Co-designing specialized retrieval with the agent preserves accuracy while halving cost.

Broader CLIO Results

Beyond agentic efficiency, CLIO achieves across controlled benchmarks, real observatory data, and distributed HPC:

- **Cross-unit retrieval:** 0.99 P@5 vs. 0.40 for BM25 on 210 synthetic documents, 80 queries across 7 domains
- **Real-world NOAA data:** +87% precision over hybrid baselines on 288 GHCN-Daily weather station records
- **Distributed scale:** p50 $<$ 42 ms across 2.5M arXiv documents on 4 NVIDIA GH200 nodes (NCSA DeltaAI), with 0.94 weak-scaling efficiency
- **IOWarp integration:** Sub-linear profiling from 17 ms (1K blobs) to 90 ms (500K blobs) on IOWarp CTE storage
- **Federation:** 1.0 routing recall across 100 namespaces while skipping 52% of irrelevant namespaces

CLIO Search

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IOWarp Community